

Detecting Early Blight Disease Severity within Tomato Plants Utilizing Machine Learning

Gerald Yu¹, John Manavalan², Muhammad Musa Qamar³, Panshul Purusharthy⁴, Rashi Kumari⁵

¹ Amador Valley High School, Pleasanton, California, United States

² Metea Valley High School, Aurora, Illinois, United States

³ Nixor College, Karachi, Pakistan

⁴ Milton Academy, Udhailiyah, Saudi Arabia

⁵ G. D. Birla Centre for Education, Kolkata, India

Abstract

Early blight in tomato plants, caused by the fungi *Alternaria solani* or *Alternaria tomatophila*, is a destructive disease that can lead to severe crop loss. This paper proposes a convolutional neural network that analyzes tomato plant leaves and detects the severity of Early Blight before pathological damage becomes irreversible. The model is trained on the PlantVillage dataset, which initially contained only Healthy and Diseased labels. HSV and Lab color segmentation were then used to estimate the percentage of leaf area affected by blight, allowing the samples to be categorized into Healthy, Mild, Moderate, and Severe. The CNN model achieved an accuracy of 86.36% and a macro F1 score of 79.64% when evaluated on a held-out test set of 330 images. The model showed the most confusion between Moderate and Mild severity reflecting the percentage-based thresholds used to define the boundary between them rather than a biological distinction. These results show that both detection and severity can be estimated directly from a photo, allowing farmers to catch the disease early, and prevent the spread of the fungus between plants.